Theo Glashier

MEng, PhD, DIC, AMIMechE

I am an early career researcher focused on promoting the resilience of society's infrastructure, notably through digitalisation. I aim to utilise both novel artificial intelligence techniques and new methods of co-production to enable new approaches for monitoring engineering structures.

Contact

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Links

Personal research website
Imperial College profile
LinkedIn

Research employment

Research Associate in Large-Scale Infrastructure Monitoring, Imperial College London
July 2025 - Present

- Designed a state-of-the-art outdoor experiment for long-term in-situ performance monitoring of traditional, asphalt, and high-strength permeable pavements
- Developed a comprehensive monitoring network for comparing the heat transfer and permeability of different pavement materials, and their evolution over multiple years
- Attained significant first-hand experience in embedding instrumentation within construction materials, while performing sensor network design, installation and commissioning
- Gained considerable research and practical knowledge at the intersection of structural engineering, hydrology and data science

Education

PhD, Civil and Environmental Engineering, Imperial College London
March 2021 - May 2025

- Thesis entitled 'Temperature-based measurement interpretation of critical civil infrastructure', doi: 10.25560/125844, and supervised by Dr Craig Buchanan
- Developed specialty expertise in data-driven techniques, statistical methods, machine learning, high performance computing and programming in Python and Matlab
- Acquired considerable experience in the long-term monitoring of infrastructure, including setting-up and troubleshooting of sensor networks and performing in-person large-scale static and dynamic testing on the operational MX3D Bridge in Amsterdam
- Gained excellent written and oral communication skills through seminars, poster and presentation competitions, teaching experiences and multiple work trips to present to academic and industrial collaborators and disseminate my research at conferences
- Co-supervised six Master students during their final-year projects, gaining significant first-hand supervision experience while leading them to publishing their work

MEng Mechanical Engineering, University of Sheffield, First-Class Honours
October 2015 - July 2019

- Thesis entitled 'Operational modal analysis for nonlinear systems', and supervised by Professor Keith Worden (grade: First)
- Undertook a third year project, equivalent to a BEng dissertation, in the design and specification of a wind turbine bearing test rig that led to a research summer internship with RGL Associates
- Developed specialty expertise in condition monitoring, structural dynamics, material integrity, solid mechanics, with grades of 89%, 82%, 80%, 75%, respectively

Non-academic appointments

E-monitoring engineer graduate scheme, Total Energies, Pau, France
October 2019 - September 2020

- Delivered long-term monitoring projects for offshore energy infrastructure, involving the large-scale testing of machines and equipment distributed across multiple international sites using datasets that were continually supplied by hundreds of sensors
- Developed excellent management skills through taking the technical lead on the CO2 emission monitoring project, building the data infrastructure template in PI System and applying it to a large portion of Total Energies' worldwide assets. I have been acknowledged for my contribution in this work, doi: 10.2118/216258-MS
- Gained invaluable experience in delivering technical presentations to internal clients

Summer internship, RGL Associates, Sheffield
Summer 2018

- Developed professional and technical experience, early in my career, by undertaking finite element (FE) analysis of wind turbine bearings and delivering projects for a small engineering firm

Skills

Programming: Python, Matlab, C, Linux command line (e.g. for High Performance Computing)

Software: Rhino, Solidworks, Labview, Ansys FE modelling and computational fluid dynamics, Microsoft Office Suite, PI System

Societal engagement: Completed Imperial's Engagement Academy 2025/2026, Great Exhibition Road Festival 2024 and 2025

Languages: English and French (both native, C2, can peer review), Italian and Spanish (A2)

Awards

- Milija Pavlovic Research Travel Fund award to attend the 11th European Workshop on Structural Health Monitoring 2024
- Imperial College General fund and Old Centralians' Trust awards for attendance at the 12th International Association on Bridge Maintenance and Safety 2024
- Awarded the 2nd Prize in the Imperial College PhD Summer Showcase 2023

- Invited to deliver an oral presentation at the 25th Young Researchers Conference 2023 at the Institution of Structural Engineers
- Awarded the Skempton PhD Scholarship

Talks

- The Centre for Infrastructure Materials Seminar Series, Department of Civil Engineering at Imperial College London, 2025
- The 12th International Association for Bridge Maintenance and Safety (IABMAS), Copenhagen, Denmark, 2024
- The 11th European Workshop on Structural Health Monitoring (EWSHM), Potsdam, Germany, 2024
- Poster presentation at Imperial College London's Department of Civil Engineering PhD conference, 2024
- Imperial College's Early Career Researcher Showcase, London, UK, 2023
- The 25th Young Researchers Conference of the Institution of Structural Engineers, London, UK, 2023
- Oral presentation at Imperial College London's Department of Civil Engineering PhD conference, 2022

Student supervision

MEng Thesis: *Analysing the thermal induced structural behaviour of the first 3D metal printed bridge*, by Maxwell Otieno Omondi, Imperial College London (2021)

MEng Thesis: *Initial investigations into pedestrian loading on the first metal 3D printed bridge*, by Austin Young, Imperial College London (2021)

MSc Thesis: *Predicting thermal-induced response of the first 3D metal printed bridge using four regression algorithms*, by Hu Zhang, Imperial College London (2021)

MEng Thesis: *Analysing the dynamic behaviour of the first metal 3D printed bridge,* by Ioan Rheinallt, Imperial College London (2022)

MSc Thesis: *Thermal response and user's behaviour modelling for MX3D - world's first metal 3D printed bridge*, by Akhil Goyal, Imperial College London (2022)

MSc Thesis: *The structural behaviour of the MX3D Bridge under dynamic and temperature effect*, by Siu Jun Lau, Imperial College London (2022)

References

Dr Rolands Kromanis, Associate Professor at the University of Twente, the Netherlands, r.kromanis@utwente.nl

Dr Thomas Reynolds, Senior Lecturer at the University of Edinburgh, t.reynolds@ed.ac.uk

Professor Chris Cheeseman, Professor of Materials Resources Engineering at Imperial College London, c.cheeseman@imperial.ac.uk

Dr Craig Buchanan, Associate Professor at Imperial College London and my PhD supervisor, craig.buchanan@imperial.ac.uk

List of publications

Journal articles

Glashier, T., Kromanis, R. and Buchanan, C. (2024) 'An iterative regression-based thermal response prediction methodology for instrumented civil infrastructure', *Advanced Engineering Informatics*, 60, p. 102347. doi: 10.1016/j.aei.2023.102347.

Glashier, T., Kromanis, R. and Buchanan, C. (2024) 'Temperature-based measurement interpretation of the MX3D Bridge', *Engineering Structures*, 305, p. 116736. doi: 10.1016/j.engstruct.2023.116736.

Reynolds, T. P. S., **Glashier, T**., Cameron, J., Kromanis, R., Zhang, P., Wynne, Z., Tessier, A., Shuldiner, A., Cammers-Goodwin, S., Vahdatikhaki, F., Nagenborg, M., Siderius, K., Buchanan, C. (2025) 'Operational data collection and analysis for a smart 3D printed footbridge', *Smart and Sustainable Built Environment,* 1-29. doi: 10.1108/SASBE-04-2025-0167

Glashier, T., Kromanis, R. and Buchanan, C. (2026) 'Preparation of long-term structural health monitoring data: Signal visualisation and processing', *Preprint to be submitted to Mechanical Systems and Signal Processing*.

Glashier, T., Kromanis, R. and Buchanan, C. (2026) 'A physics-informed, data-driven temperature-based interpretation of the in-service MX3D Bridge', *Preprint to be submitted to Structural Health Monitoring*.

Goyal A., **Glashier, T.**, and Buchanan, C. (2026) 'Sensor fusion for load mapping on bridges', *Preprint to be submitted to Engineering Structures*.

Conference papers

Glashier, T. (2023) 'Predicting the environmental response of critical infrastructure, using the first metal 3D printed structure as a case study', in *Proceedings of the 25th Young Researchers Conference. Institution of Structural Engineers*, 62–63. Available at: https://www.istructe.org/getattachment/Resources/Training/Young-researchers-conference-2023/Young-researchers-conference-2023-Proceedings.pdf?lang=en-GB.

Glashier, T., Kromanis, R. and Buchanan, C. (2024) 'Temperature-based damage detection for the commissioning dataset of the MX3D Bridge', in *Proceedings of the 11th European Workshop on Structural Health Monitoring (EWSHM 2024), June 10-13, 2024*. Potsdam, Germany: e-Journal of Nondestructive Testing. doi: 10.58286/29866.

Book chapters

Glashier, T., Buchanan, C. and Kromanis, R. (2024) 'Thermal response prediction of the MX3D bridge's operational dataset', in Jensen, J. S., Frangopol, D. M., and Schmidt, J. W. (eds) Bridge Maintenance, Safety, Management, Digitalization and Sustainability. London: CRC Press, pp. 2511–2519. doi: 10.1201/9781003483755-299.

Archives

Cameron, J., Tessier, A., Buchanan, C., Wynne, Z., Reynolds, T., **Glashier, T.,** Kromanis, R., Cammers-Goodwin, S., Bibliowicz, J. and Shuldiner, A. (2024) 'Data Archive from the MX3D Bridge in Amsterdam.', Available at: https://zenodo.org/records/10641551.